

Supplemental Appendix: New Evidence of the Usefulness of Real-Time Sentiment Indices in Econometric and Judgement Based Forecasting Models(Not for Publication)*

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Abstract

This supplemental appendix contains additional tables to accompany the empirical results reported in the main paper.

Keywords: Judgement, Real-time forecasting, Forecast Rationality and Efficiency, Narrative and Consumer Sentiment, Large Language Models, Text-as-Data.

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1 Supplemental Tables

Table 1: Summary of Efficiency Test Results with LLM-based Narrative Sentiment and Macroeconomic Variables - Subsample 2011Q1–2019Q4*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	NO	NO	NO	NO	0.6135	0.5689	0.5567	0.9012
h=2	NO	NO	NO	NO	0.5365	0.1580	0.3264	0.4873
h=3	NO	NO	NO	NO	0.2384	0.8829	0.7268	0.2597
h=4	NO	NO	NO	NO	0.1198	0.2893	0.4925	0.1429
<i>CPI Inflation</i>								
h=1	YES	NO	NO	YES	0.0016	0.2908	0.3543	0.0613
h=2	YES	YES	YES	NO	0.0048	0.0379	0.0250	0.1047
h=3	YES	NO	NO	YES	0.0007	0.5114	0.6166	0.0168
h=4	YES	NO	NO	NO	0.0866	0.8458	0.7266	0.2645
<i>3-Month Treasury Bill Rate</i>								
h=1	NO	NO	NO	NO	0.3950	0.4686	0.4284	0.4253
h=2	YES	NO	NO	YES	0.0178	0.4825	0.4529	0.0421
h=3	YES	NO	NO	YES	0.0011	0.5201	0.5248	0.0011
h=4	YES	NO	NO	YES	0.0003	0.4801	0.5418	0.0001
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	NO	NO	YES	0.0006	0.6557	0.5922	0.0112
h=2	YES	NO	NO	YES	0.0000	0.4202	0.4812	0.0081
h=3	YES	NO	YES	YES	0.0000	0.5037	0.0868	0.0021
h=4	YES	NO	YES	YES	0.0000	0.1263	0.0386	0.0001
<i>Unemployment</i>								
h=1	YES	NO	NO	YES	0.0041	0.9939	0.8661	0.0167
h=2	NO	NO	NO	NO	0.6828	0.2115	0.6889	0.5917
h=3	YES	NO	NO	NO	0.0584	0.1883	0.6137	0.1518
h=4	NO	NO	NO	NO	0.1030	0.3136	0.7039	0.1778

*Notes: See notes to Table 7 in the main paper. Sample: 2011Q1–2019Q4.

Table 2: Summary of Efficiency Test Results with University of Michigan and Conference Board Sentiment and Macroeconomic Variables - Subsample 2011Q1–2019Q4*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	NO	NO	NO	NO	0.8661	0.9526	0.9002	0.8200
h=2	NO	NO	NO	NO	0.6057	0.2400	0.6583	0.6816
h=3	NO	NO	NO	NO	0.3078	0.5267	0.6903	0.2106
h=4	NO	NO	NO	NO	0.2022	0.5312	0.8514	0.2972
<i>CPI Inflation</i>								
h=1	YES	NO	NO	YES	0.0043	0.3429	0.3763	0.0009
h=2	NO	NO	NO	NO	0.5621	0.4895	0.7903	0.7874
h=3	YES	NO	NO	NO	0.0043	0.9164	0.3972	0.1081
h=4	YES	NO	NO	YES	0.0316	0.3618	0.4546	0.0575
<i>3-Month Treasury Bill Rate</i>								
h=1	NO	NO	NO	NO	0.2470	0.5962	0.7939	0.2851
h=2	YES	NO	NO	YES	0.0010	0.1948	0.3671	0.0509
h=3	YES	NO	NO	YES	0.0000	0.4297	0.6929	0.0582
h=4	YES	NO	NO	YES	0.0000	0.4140	0.5228	0.0080
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	NO	NO	YES	0.0018	0.9397	0.6490	0.0086
h=2	YES	NO	NO	YES	0.0000	0.5484	0.2605	0.0004
h=3	YES	NO	NO	YES	0.0000	0.5254	0.2849	0.0024
h=4	YES	NO	NO	YES	0.0000	0.5119	0.3483	0.0000
<i>Unemployment</i>								
h=1	YES	NO	NO	NO	0.0082	0.3582	0.5626	0.7607
h=2	NO	NO	NO	NO	0.1814	0.7434	0.6587	0.8788
h=3	NO	NO	NO	NO	0.2660	0.4299	0.6379	0.5243
h=4	NO	NO	NO	NO	0.1126	0.9548	0.6004	0.3468

*Notes: See notes to Table 8 in the main paper. Sample: 2011Q1–2019Q4.

Table 3: Summary of Efficiency Test Results with LLM-based Narrative Sentiment and Macroeconomic Variables - Subsample 2020Q1–2024Q1*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	YES	YES	YES	YES	0.0000	0.0697	0.0491	0.0003
h=2	YES	NO	NO	NO	0.0415	0.7855	0.1385	0.4135
h=3	YES	NO	YES	YES	0.0017	0.2892	0.0574	0.0689
h=4	YES	NO	YES	YES	0.0001	0.5895	0.0020	0.0007
<i>CPI Inflation</i>								
h=1	YES	NO	NO	NO	0.0188	0.2232	0.2377	0.3441
h=2	YES	YES	NO	YES	0.0011	0.0962	0.5489	0.0035
h=3	YES	YES	YES	YES	0.0016	0.0974	0.0299	0.0127
h=4	YES	NO	YES	YES	0.0000	0.3119	0.0727	0.0671
<i>3-Month Treasury Bill Rate</i>								
h=1	YES	YES	NO	YES	0.0513	0.0424	0.1260	0.0244
h=2	YES	NO	YES	YES	0.0013	0.1137	0.0261	0.0111
h=3	YES	NO	NO	YES	0.0004	0.8896	0.2322	0.0175
h=4	YES	NO	NO	YES	0.0000	0.4371	0.5097	0.0005
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	NO	NO	YES	0.0032	0.4507	0.3613	0.0146
h=2	YES	NO	NO	YES	0.0051	0.5221	0.1588	0.0356
h=3	YES	NO	NO	YES	0.0005	0.5118	0.3647	0.0145
h=4	YES	NO	NO	YES	0.0000	0.3833	0.5235	0.0772
<i>Unemployment</i>								
h=1	YES	NO	YES	YES	0.0000	0.1925	0.0182	0.0011
h=2	YES	NO	YES	YES	0.0000	0.2112	0.0219	0.0000
h=3	YES	NO	YES	YES	0.0000	0.5134	0.0966	0.0005
h=4	YES	NO	YES	YES	0.0000	0.8146	0.0906	0.0012

*Notes: See notes to Table 7 in the main paper. Sample: 2020Q1–2024Q1.

Table 4: Summary of Efficiency Test Results with University of Michigan and Conference Board Sentiment and Macroeconomic Variables - Subsample 2020Q1–2024Q1*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	YES	YES	NO	YES	0.0007	0.0420	0.8168	0.0003
h=2	YES	NO	NO	YES	0.0130	0.5366	0.2997	0.0871
h=3	NO	NO	NO	NO	0.1082	0.4791	0.7069	0.2344
h=4	YES	YES	YES	YES	0.0000	0.0001	0.0000	0.0675
<i>CPI Inflation</i>								
h=1	YES	NO	NO	NO	0.0311	0.4456	0.1046	0.1199
h=2	YES	NO	NO	YES	0.0262	0.3655	0.5996	0.0429
h=3	YES	YES	YES	YES	0.0001	0.0061	0.0056	0.0264
h=4	YES	YES	YES	YES	0.0000	0.0606	0.0808	0.0021
<i>3-Month Treasury Bill Rate</i>								
h=1	YES	YES	YES	YES	0.0085	0.0065	0.0134	0.0230
h=2	YES	YES	YES	YES	0.0000	0.0483	0.0067	0.0214
h=3	YES	NO	YES	YES	0.0004	0.1531	0.0177	0.0038
h=4	YES	NO	NO	YES	0.0000	0.3977	0.1167	0.0001
<i>10-Year Treasury Bond Rate</i>								
h=1	NO	YES	NO	YES	0.1024	0.0872	0.5322	0.0505
h=2	YES	YES	YES	YES	0.0002	0.0456	0.0177	0.0035
h=3	YES	NO	NO	YES	0.0000	0.1413	0.1599	0.0001
h=4	YES	NO	YES	YES	0.0000	0.7127	0.0103	0.0010
<i>Unemployment</i>								
h=1	YES	YES	YES	YES	0.0000	0.0033	0.0437	0.0023
h=2	YES	YES	NO	YES	0.0000	0.0417	0.2404	0.0003
h=3	YES	YES	NO	YES	0.0000	0.0690	0.1315	0.0000
h=4	YES	YES	YES	YES	0.0000	0.0781	0.0465	0.0000

*Notes: See notes to Table 8 in the main paper. Sample: 2020Q1–2024Q1.

Table 5: Summary of Efficiency Test Results with LLM-based Narrative Sentiment and Macroeconomic Variables - Philadelphia Fed Realization - Full Sample*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	NO	NO	NO	NO	0.4629	0.3819	0.4193	0.4625
h=2	NO	NO	NO	NO	0.9218	0.7407	0.4371	0.7771
h=3	NO	NO	NO	NO	0.8950	0.9418	0.7785	0.9491
h=4	NO	NO	NO	NO	0.2123	0.6580	0.4997	0.9558
<i>CPI Inflation</i>								
h=1	YES	NO	NO	NO	0.0377	0.8564	0.5636	0.1199
h=2	YES	NO	YES	YES	0.0022	0.7938	0.0707	0.0056
h=3	YES	NO	NO	NO	0.0283	0.1845	0.1919	0.1786
h=4	YES	NO	NO	NO	0.0000	0.8337	0.1755	0.1069
<i>3-Month Treasury Bill Rate</i>								
h=1	YES	NO	NO	YES	0.0211	0.7103	0.4899	0.0299
h=2	NO	NO	YES	NO	0.3119	0.5822	0.0982	0.2519
h=3	YES	NO	YES	NO	0.0599	0.8459	0.0249	0.1783
h=4	YES	NO	YES	YES	0.0099	0.9294	0.0539	0.0691
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	NO	NO	YES	0.0026	0.8878	0.7945	0.0280
h=2	YES	NO	NO	YES	0.0000	0.5994	0.2700	0.0002
h=3	YES	NO	NO	YES	0.0001	0.9349	0.7345	0.0014
h=4	YES	NO	NO	YES	0.0001	0.9929	0.7315	0.0035
<i>Unemployment</i>								
h=1	YES	YES	NO	YES	0.0000	0.0672	0.1435	0.0001
h=2	NO	NO	NO	YES	0.3087	0.6342	0.1484	0.0965
h=3	YES	NO	NO	YES	0.0026	0.5101	0.2290	0.0138
h=4	NO	NO	NO	NO	0.1307	0.3112	0.1186	0.2855

*Notes: See notes to Table 7 in the main paper. Sample: 2011Q1–2024Q1. Forecast errors use the Philadelphia Fed official Realization benchmark.

Table 6: Summary of Efficiency Test Results with University of Michigan and Conference Board Sentiment and Macroeconomic Variables - Philadelphia Fed Realization - Full Sample*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	NO	YES	YES	NO	0.2987	0.0659	0.0936	0.2824
h=2	NO	NO	NO	NO	0.9358	0.2444	0.4281	0.8303
h=3	NO	NO	NO	NO	0.5245	0.2092	0.3372	0.4575
h=4	NO	NO	NO	NO	0.7055	0.6986	0.7337	0.6015
<i>CPI Inflation</i>								
h=1	YES	NO	YES	YES	0.0023	0.4283	0.0101	0.0012
h=2	YES	NO	YES	YES	0.0202	0.1389	0.0961	0.0315
h=3	YES	NO	YES	YES	0.0121	0.7763	0.0200	0.0187
h=4	YES	NO	YES	YES	0.0000	0.2689	0.0090	0.0000
<i>3-Month Treasury Bill Rate</i>								
h=1	YES	NO	NO	YES	0.0160	0.1910	0.4534	0.0094
h=2	YES	YES	YES	YES	0.0014	0.0113	0.0146	0.0120
h=3	YES	YES	YES	YES	0.0000	0.0045	0.0003	0.0017
h=4	YES	YES	YES	YES	0.0000	0.0017	0.0001	0.0002
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	YES	YES	YES	0.0004	0.0075	0.0971	0.0052
h=2	YES	YES	YES	YES	0.0001	0.0155	0.0764	0.0006
h=3	YES	YES	YES	YES	0.0000	0.0120	0.0360	0.0039
h=4	YES	YES	YES	YES	0.0000	0.0038	0.0183	0.0312
<i>Unemployment</i>								
h=1	YES	YES	YES	YES	0.0000	0.0982	0.0551	0.0000
h=2	YES	NO	NO	YES	0.0001	0.2532	0.1965	0.0000
h=3	YES	NO	NO	YES	0.0001	0.5408	0.1458	0.0003
h=4	NO	NO	NO	NO	0.4596	0.2929	0.1689	0.2941

*Notes: See notes to Table 8 in the main paper. Sample: 2011Q1–2024Q1. Forecast errors use the Philadelphia Fed official Realization benchmark.

Table 7: Summary of Efficiency Test Results with LLM-based Narrative Sentiment and Macroeconomic Variables - Philadelphia Fed Realization - Subsample 2011Q1–2019Q4*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	NO	NO	NO	NO	0.3661	0.9220	0.7903	0.9061
h=2	NO	NO	NO	NO	0.3880	0.2194	0.6220	0.6077
h=3	NO	NO	NO	NO	0.8365	0.4960	0.8025	0.8278
h=4	NO	NO	NO	NO	0.1321	0.9143	0.7957	0.6656
<i>CPI Inflation</i>								
h=1	NO	NO	NO	NO	0.9411	0.6106	0.7943	0.9770
h=2	YES	YES	YES	YES	0.0119	0.0010	0.0088	0.0071
h=3	NO	NO	NO	NO	0.3771	0.6819	0.8292	0.3073
h=4	YES	NO	NO	YES	0.0066	0.8784	0.9721	0.0764
<i>3-Month Treasury Bill Rate</i>								
h=1	YES	NO	NO	NO	0.0188	0.4721	0.2521	0.4779
h=2	YES	NO	NO	YES	0.0069	0.3968	0.4048	0.0328
h=3	YES	NO	NO	YES	0.0000	0.5343	0.5002	0.0001
h=4	YES	NO	NO	YES	0.0000	0.8835	0.6193	0.0000
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	NO	NO	YES	0.0001	0.5982	0.6874	0.0029
h=2	YES	NO	NO	YES	0.0000	0.7270	0.6527	0.0070
h=3	YES	NO	NO	YES	0.0000	0.4376	0.2707	0.0015
h=4	YES	NO	YES	YES	0.0000	0.3108	0.0200	0.0001
<i>Unemployment</i>								
h=1	YES	NO	NO	YES	0.0347	0.7605	0.7781	0.0717
h=2	NO	NO	NO	NO	0.8926	0.2296	0.7055	0.6881
h=3	NO	NO	NO	NO	0.3384	0.1808	0.6005	0.3775
h=4	YES	NO	NO	NO	0.0969	0.2942	0.6053	0.2539

*Notes: See notes to Table 7 in the main paper. Sample: 2011Q1–2019Q4. Forecast errors use the Philadelphia Fed official Realization benchmark.

Table 8: Summary of Efficiency Test Results with University of Michigan and Conference Board Sentiment and Macroeconomic Variables - Philadelphia Fed Realization - Subsample 2011Q1–2019Q4*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	NO	NO	NO	NO	0.7481	0.9083	0.9364	0.7522
h=2	NO	NO	NO	NO	0.3139	0.4528	0.8734	0.4380
h=3	NO	NO	NO	NO	0.5185	0.3988	0.3758	0.6351
h=4	NO	NO	NO	NO	0.1879	0.4589	0.7751	0.1529
<i>CPI Inflation</i>								
h=1	NO	NO	NO	NO	0.9787	0.8637	0.9667	0.8818
h=2	NO	NO	NO	NO	0.6455	0.8363	0.9859	0.9986
h=3	NO	NO	NO	NO	0.1024	0.5095	0.4883	0.2076
h=4	YES	NO	NO	YES	0.0011	0.5197	0.7021	0.0012
<i>3-Month Treasury Bill Rate</i>								
h=1	YES	NO	NO	NO	0.0216	0.5149	0.7491	0.2265
h=2	YES	NO	NO	YES	0.0000	0.1715	0.4109	0.0174
h=3	YES	NO	NO	YES	0.0000	0.2415	0.4713	0.0212
h=4	YES	NO	NO	YES	0.0000	0.3790	0.6217	0.0203
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	NO	NO	YES	0.0002	0.9074	0.6383	0.0032
h=2	YES	NO	NO	YES	0.0000	0.5075	0.3101	0.0021
h=3	YES	NO	NO	YES	0.0000	0.3972	0.2026	0.0008
h=4	YES	NO	NO	YES	0.0000	0.3444	0.2059	0.0005
<i>Unemployment</i>								
h=1	YES	NO	NO	NO	0.0285	0.2863	0.5333	0.9035
h=2	NO	NO	NO	NO	0.7128	0.9371	0.6242	0.9159
h=3	NO	NO	NO	NO	0.6929	0.4015	0.6463	0.7009
h=4	NO	NO	NO	NO	0.2549	0.8792	0.5897	0.3459

*Notes: See notes to Table 8 in the main paper. Sample: 2011Q1–2019Q4. Forecast errors use the Philadelphia Fed official Realization benchmark.

Table 9: Summary of Efficiency Test Results with LLM-based Narrative Sentiment and Macroeconomic Variables - Philadelphia Fed Realization - Subsample 2020Q1–2024Q1*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	YES	NO	NO	YES	0.0897	0.1271	0.3110	0.0758
h=2	YES	NO	NO	YES	0.0010	0.1393	0.1271	0.0377
h=3	YES	NO	YES	YES	0.0003	0.2077	0.0205	0.0315
h=4	YES	NO	YES	YES	0.0487	0.2905	0.0494	0.0289
<i>CPI Inflation</i>								
h=1	YES	NO	YES	NO	0.0028	0.6100	0.0902	0.4371
h=2	YES	YES	NO	YES	0.0008	0.0142	0.2465	0.0121
h=3	YES	YES	YES	YES	0.0000	0.0353	0.0217	0.0068
h=4	YES	NO	YES	YES	0.0000	0.5620	0.0137	0.0476
<i>3-Month Treasury Bill Rate</i>								
h=1	YES	NO	NO	YES	0.0071	0.2123	0.5017	0.0042
h=2	YES	YES	YES	YES	0.0008	0.0531	0.0159	0.0068
h=3	YES	NO	NO	YES	0.0008	0.7287	0.1485	0.0144
h=4	YES	NO	NO	YES	0.0000	0.5008	0.4973	0.0009
<i>10-Year Treasury Bond Rate</i>								
h=1	YES	NO	NO	YES	0.0516	0.6530	0.6233	0.0967
h=2	YES	NO	NO	YES	0.0029	0.6896	0.7057	0.0449
h=3	YES	NO	YES	YES	0.0007	0.7744	0.0507	0.0171
h=4	YES	NO	NO	YES	0.0001	0.4206	0.5676	0.0100
<i>Unemployment</i>								
h=1	YES	NO	YES	YES	0.0000	0.2081	0.0231	0.0021
h=2	YES	NO	YES	YES	0.0000	0.3004	0.0023	0.0000
h=3	YES	NO	YES	YES	0.0000	0.4051	0.0237	0.0002
h=4	YES	NO	YES	YES	0.0000	0.8215	0.0632	0.0009

*Notes: See notes to Table 7 in the main paper. Sample: 2020Q1–2024Q1. Forecast errors use the Philadelphia Fed official Realization benchmark.

Table 10: Summary of Efficiency Test Results with University of Michigan and Conference Board Sentiment and Macroeconomic Variables - Philadelphia Fed Realization - Subsample 2020Q1–2024Q1*

h	Hypothesis				p-value			
	(1) $H_0 : \alpha = \beta = \gamma = 0$	(2) $H_0 : \alpha = 0$	(3) $H_0 : \beta = 0$	(4) $H_0 : \gamma = 0$	(1)	(2)	(3)	(4)
<i>Industrial Production</i>								
h=1	YES	NO	NO	YES	0.0004	0.5894	0.2136	0.0014
h=2	YES	YES	NO	YES	0.0052	0.0012	0.2494	0.0020
h=3	YES	NO	NO	YES	0.0046	0.1757	0.2839	0.0146
h=4	YES	YES	YES	NO	0.0043	0.0014	0.0016	0.3005
<i>CPI Inflation</i>								
h=1	YES	YES	YES	YES	0.0181	0.0748	0.0442	0.0558
h=2	YES	NO	YES	YES	0.0002	0.4206	0.0645	0.0267
h=3	YES	YES	YES	YES	0.0000	0.0161	0.0116	0.0020
h=4	YES	YES	YES	YES	0.0000	0.0043	0.0039	0.0000
<i>3-Month Treasury Bill Rate</i>								
h=1	YES	YES	YES	YES	0.0002	0.0147	0.0623	0.0001
h=2	YES	YES	YES	YES	0.0000	0.0425	0.0178	0.0251
h=3	YES	NO	YES	YES	0.0002	0.1662	0.0119	0.0097
h=4	YES	NO	YES	YES	0.0000	0.2825	0.0989	0.0002
<i>10-Year Treasury Bond Rate</i>								
h=1	NO	NO	NO	YES	0.1471	0.1228	0.7045	0.0826
h=2	YES	YES	YES	YES	0.0005	0.0962	0.0544	0.0034
h=3	YES	YES	YES	YES	0.0000	0.0465	0.0413	0.0049
h=4	YES	NO	YES	YES	0.0000	0.5825	0.0521	0.0007
<i>Unemployment</i>								
h=1	YES	YES	YES	YES	0.0000	0.0030	0.0307	0.0093
h=2	YES	YES	YES	YES	0.0000	0.0193	0.0683	0.0001
h=3	YES	YES	NO	YES	0.0000	0.0489	0.1486	0.0000
h=4	YES	YES	YES	YES	0.0000	0.0904	0.0532	0.0000

*Notes: See notes to Table 8 in the main paper. Sample: 2020Q1–2024Q1. Forecast errors use the Philadelphia Fed official Realization benchmark.